# Custom-HBM Array

## LLM Decode Accelerator

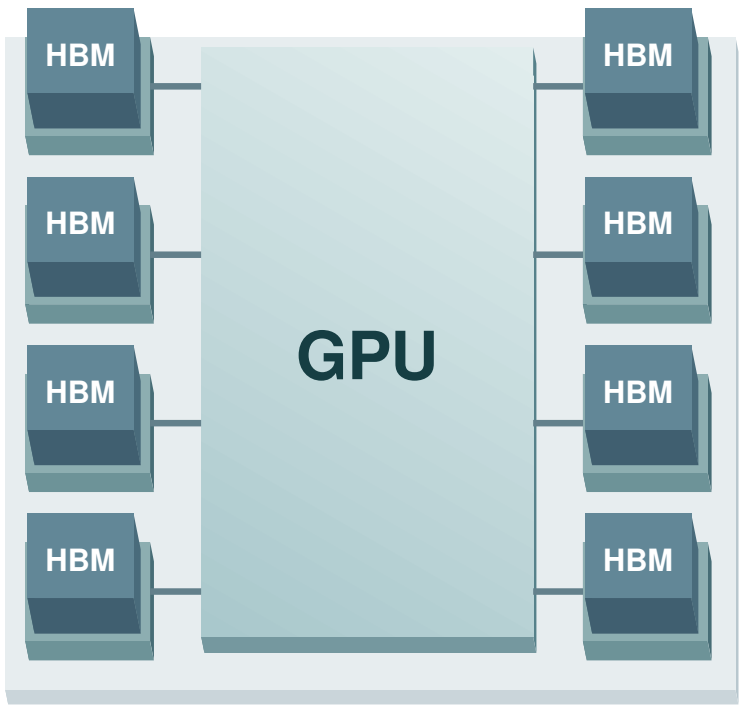
Paper

e-Brochure

### The decoding bottleneck

Memory-bound LLM decoding leaves traditional GPU compute underutilized, while HBM bandwidth is difficult to scale.

**Conventional GPU + HBM**

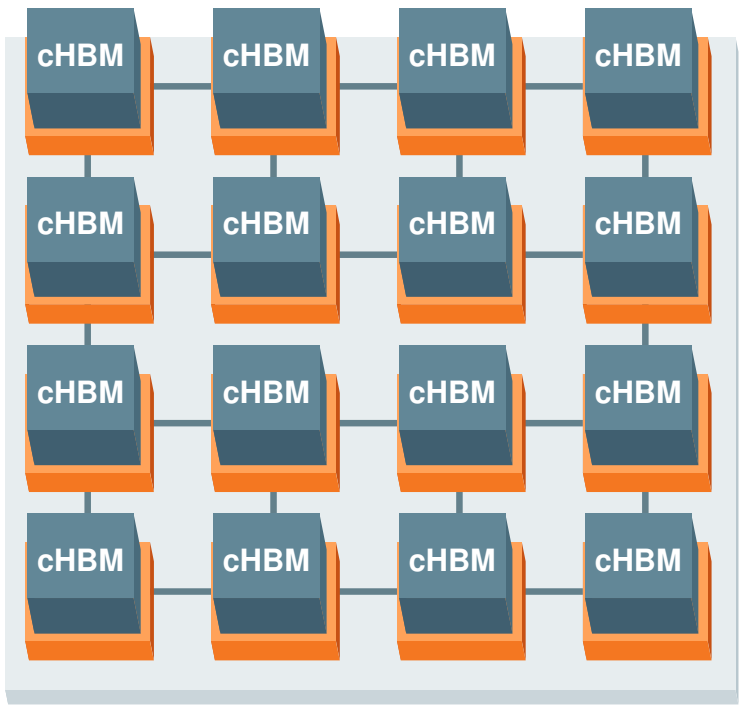



High memory use, low compute utilization

### Scalable cHBM chiplets

Each cHBM chiplet integrates an NPU into its logic die for LLM decoding. The array scales memory bandwidth to match workload needs.

**4 × 4 cHBM Array**

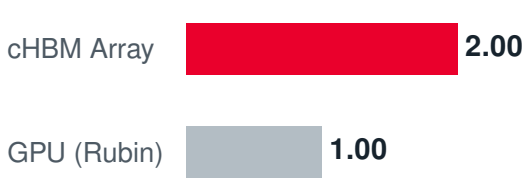



Hybrid tensor and pipeline parallelism

## Performance

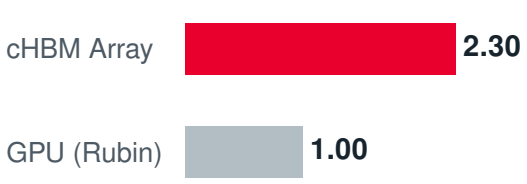
**2.00×**
Memory bandwidth

**Memory bandwidth**




UP TO
**2.30×**
Decode throughput

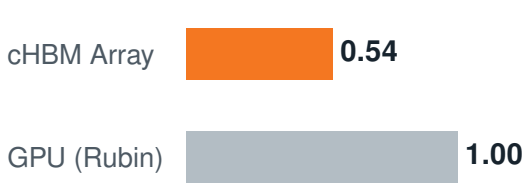
**Tokens per second**




## Energy efficiency

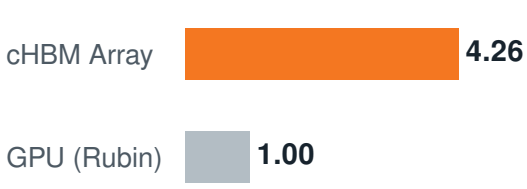
**–46%**
Power consumption

**Power consumption**




UP TO
**4.26×**
Performance per watt

**Tokens per second / watt**




Relative to GPU (Rubin). Chart values normalize GPU (Rubin) to 1.00.